Thiranex Internship

Task 4: Data Cleaning & Reporting Automation

Objective: Automate a repeatable workflow to clean sales data, handle missing values and duplicates, standardize formats, calculate summary metrics, and generate reports.

1. Dataset Overview

The synthetic demonstration dataset has 332 raw rows, including deliberately introduced duplicates, missing values, inconsistent region capitalization, and invalid dates.

2. Cleaning Workflow

• Load the raw CSV and preserve the original.
• Remove exact duplicate rows.
• Standardize text and region labels.
• Convert date and numeric columns.
• Fill missing text with Unknown, numeric values with the median, and dates with the median valid date.
• Calculate revenue as Quantity × Unit Price.
• Export cleaned data and an automated quality report.

3. Results

Rows before: 332
Duplicates removed: 12
Rows after deduplication: 320
Missing cells before imputation: 95
Missing cells after cleaning: 0
Total revenue: ■1,034,822.47
Top region: West
Top product: Laptop

4. Recommendations

Keep raw data unchanged, document imputation rules, validate suspicious values, and review the quality report before distributing the output. Replace the synthetic data with a dataset provided by Thiranex if available.

Data Note

No actual dataset was attached to the task screenshot. The data is synthetic and intended for demonstration only.